University of Economics and Business
Vietnam Nationtal University

Final Assignment

Topic: Analyzing business data of the e-commerce platform Olist

Course: Data Analysis in Business and Economics

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LỜI CAM ĐOAN

Tôi xin cam đoan đề tài tiểu luận kết thúc học phần " Phân tích dữ liệu trong kinh tế và kinh doanh" là của tác giả trực tiếp nghiên cứu, thực hiện. Tôi xin cam đoan những vấn đề được trình bày trong bài tiểu luận, các số liệu, kết quả và các dẫn chứng là do tôi tự tìm hiểu, có sự tham khảo, sưu tầm kế thừa từ những nghiên cứu của các tác giả đi trước. Các số liệu và kết quả nghiên cứu là trung thực, có trích dẫn rõ ràng.

Tôi xin hoàn toàn chịu trách nhiệm về những cam kết này

LỜI CẢM ƠN

Trong suốt quá trình học tập tại Trường Đại học Kinh tế - Đại học quốc gia Hà Nội , em đã nhận được rất nhiều sự quan tâm, hướng dẫn và hỗ trợ từ quý thầy cô, bạn bè và Nhà trường. Đây chính là nguồn động lực lớn giúp em có thêm kiến thức và sự tự tin để hoàn thành bài báo cáo môn Phân tích dữ liệu trong kinh tế và kinh doanh này. Trước hết, em xin bày tỏ lòng biết ơn chân thành đến cô Trần Thị Hoa Thơm và thầy Vũ Văn Hưởng, giảng viên hướng dẫn môn học, người đã luôn tận tâm giảng dạy, chia sẻ những kiến thức thực tế quý báu về lĩnh vực Phân tích dữ liệu . Sự tận tụy và nhiệt huyết của cô đã giúp em hiểu rõhơn về vai trò của dữ liệu trong việc hỗ trợ hoạt động kinh doanh hiện đại. Ngoài ra, em xin cảm ơn Trường Đại học Kinh tế đã đầu tư cơ sở vật chất hiện đại, tài liệu học tập phong phú, góp phần hỗ trợ rất nhiều cho quá trình rèn luyện và phát triển kỹ năng của sinh viên. Mặc dù đã cố gắng nỗ lực hoàn thiện bài báo cáo, nhưng với giới hạn về kinh nghiệm thực tiễn và khả năng nghiên cứu còn khiêm tốn, em nhận thấy bài làm của mình vẫn còn nhiều thiếu sót. Em rất mong nhận được những ý kiến đóng góp, nhận xét từ quý thầy cô để em có thể rút ra bài học kinh nghiệm cho bản thân trong quá trình học tập và công tác sau này. Một lần nữa, em xin chân thành cảm ơn cô Trần Thị Hoa Thơm và thầy Vũ Văn Hưởng đã tạo điều kiện thuận lợi cho em trong suốt quá trình học tập và thực hiện bài báo cáo này.

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List of abbreviations	
SMBs	medium-sized businesses
ERP	Enterprise Resource Planning
GMV	Gross Merchandise Volume

1. Introduction

1.1 Reasons for choosing the topic

In the context of the ongoing global digital transformation, e-commerce has become an inevitable trend and a major driving force behind the development of the digital economy. Selling on e-commerce platforms not only helps businesses expand their markets and reach millions of potential customers but also optimizes operating costs and improves business efficiency through the application of technology and data analysis. In particular, marketplace platforms have enabled small and medium-sized enterprises (SMEs) to compete fairly with large corporations, contributing to the promotion of entrepreneurship and comprehensive economic development. This report focuses on leveraging the publicly available dataset of Olist – the largest online store management platform in the Brazilian market. With over 100,000 orders processed across various e-commerce platforms from 2016 to 2018, the dataset provides a comprehensive view of the entire transaction journey: from customer placement, payment process, seller shipping performance, to post-purchase feedback. Through the application of Power BI tools and data query languages, the study will model core business metrics, thereby identifying bottlenecks in the supply chain and discovering patterns in consumer behavior in one of the most dynamic e-commerce markets in South America.

1.2 About Olist

Olist is a Brazilian e-commerce technology company founded in 2015 by Tiago Dalvi and headquartered in Curitiba. It operates primarily as a "marketplace of marketplaces," acting as an integrator that connects small and medium-sized businesses (SMBs) to the largest e-commerce platforms in Brazil.

Olist's core mission is to empower entrepreneurship by providing a complete ecosystem for sellers. Instead of managing individual accounts on various sites, a seller can sign a single contract with Olist to have their products listed on major marketplaces like Mercado Libre, Amazon, and B2W.

The company has evolved into a comprehensive operating system for retail, offering several integrated solutions:

• Olist Store: The main platform that manages product advertisements and listings across up to 13 different marketplaces.

• Tiny ERP: A centralized management system for inventory, orders, and financial invoicing acquired to streamline operations.

• Olist Pax: A logistics and fulfillment service that provides faster shipping through optimized routes and partnerships with carriers.
• Fintech Services: High-margin financial products, such as working capital loans and payment processing, to support SMB growth.
As of recent data, Olist has a network of over 100,000 SMB sellers, features more than 500,000 registered products, and has processed over $1.5 billion in Gross Merchandise Volume (GMV).
(Retail et al., 2021)

1.3 The Purchasing and Order Process

The process of buying and fulfilling an order through Olist involves several key stages between the marketplace, Olist, the seller, and the customer:

- Product Listing: Olist acts as the official store within major marketplaces; sellers register their products and set competitive prices via the Olist platform.
- Customer Transaction: An end-customer visits a partner marketplace (e.g., Mercado Libre) and purchases a product listed by Olist.
- Order Notification: The marketplace notifies Olist of the purchase. One order may contain multiple items belonging to different independent sellers.
- Payment and Creation: Once the payment is authorized, Olist creates a specific order for the respective seller to fulfill.
- Fulfillment and Logistics: The seller is notified to prepare the order. Olist manages the logistics—including generating shipments and postal documents—working with its logistics partners to deliver the goods directly to the customer.
- Order Status Tracking: The order moves through a standard flow: Approved (payment authorized) → Invoiced (packages created) → Shipped → Delivered.
- Customer Feedback: After the product is received or the estimated delivery date is reached, the customer receives a satisfaction survey via email to rate their experience.

1.4 Project objective

The objective of this report is to analyze Olist's business operations to identify the problems it is facing, and then propose solutions to address these problems and increase revenue through 3 groups of factors: products, customers, sellers

2. Content

2.1 Introduction to the dataset

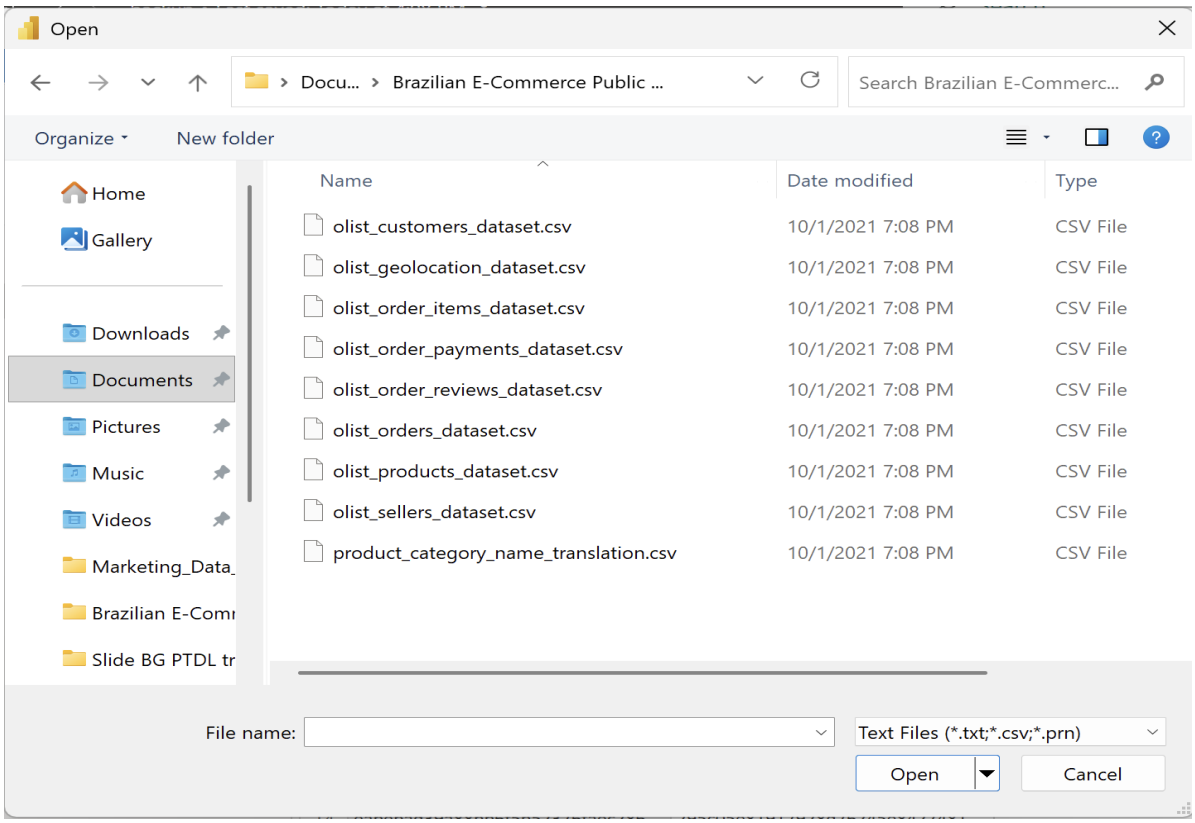
This is a Brazilian ecommerce public dataset of orders made at Olist Store. The dataset has information of 100k orders from 2016 to 2018 made at multiple marketplaces in Brazil. Its features allows viewing an order from multiple dimensions: from order status, price, payment and freight performance to customer location, product attributes and finally reviews written by customers.
The dataset include: (Orders_dataset, item_dataset, customer_dataset, geolocation_dataset, review_dataset, product_dataset, seller_dataset, payments_dataset, product_caterogy_name_translation)

- Orders (order_id, customer_id, order_status, order_purchase_timestamp, order_approved_at, order_delivered_carrier_date, order_delivered_customer_date, order_estimated_delivery_date). Orders are distinguished from each other by their order_id. Each order's delivery status and shipping progress are also updated and tracked.
- Order Items (order_id, order_item_id, product_id, seller_id, shipping_limit_date, price, freight_value). Orders are distinguished by their order_id. Products are distinguished by their order_item_id. An order_id can have multiple order_item_ids, and each order_item_id belongs to only one order_id. Additionally, each order_item_id belongs to a product_id. When a customer purchases the same product in large quantities, there will be multiple order_item_ids but only one product_id. Each order_item_id also determines the maximum delivery time, value, and shipping cost for each specific order_item_id.
- Products (product_id, product_category_name, product_name_length, product_desciption_length, product_photos qty, product_weight_g, product_length_cm, product_height_cm, product_width_cm).
- Sellers (seller_id, seller_city, seller_state, seller_zip_code_prefix).
- Customers (customer_id, customer_unique_id, customer_zip_code_prefix, customer_citycustomer_state)
- Geolocation (geolocation_zip_code_prefix, geolocation_lat, geolocation_lng, geolocation_city, geolocation_state)

- Payments (order_id, payment_sequential, payment_type, payment_installments, payment_value)
- Reviews (review_id, order_id, review_score, review_comment_title, review_comment_message, review_creation_date, review_answer_timestamp)
- Product translation (product_category_name, product_category_name_english)

2.2 Data processing

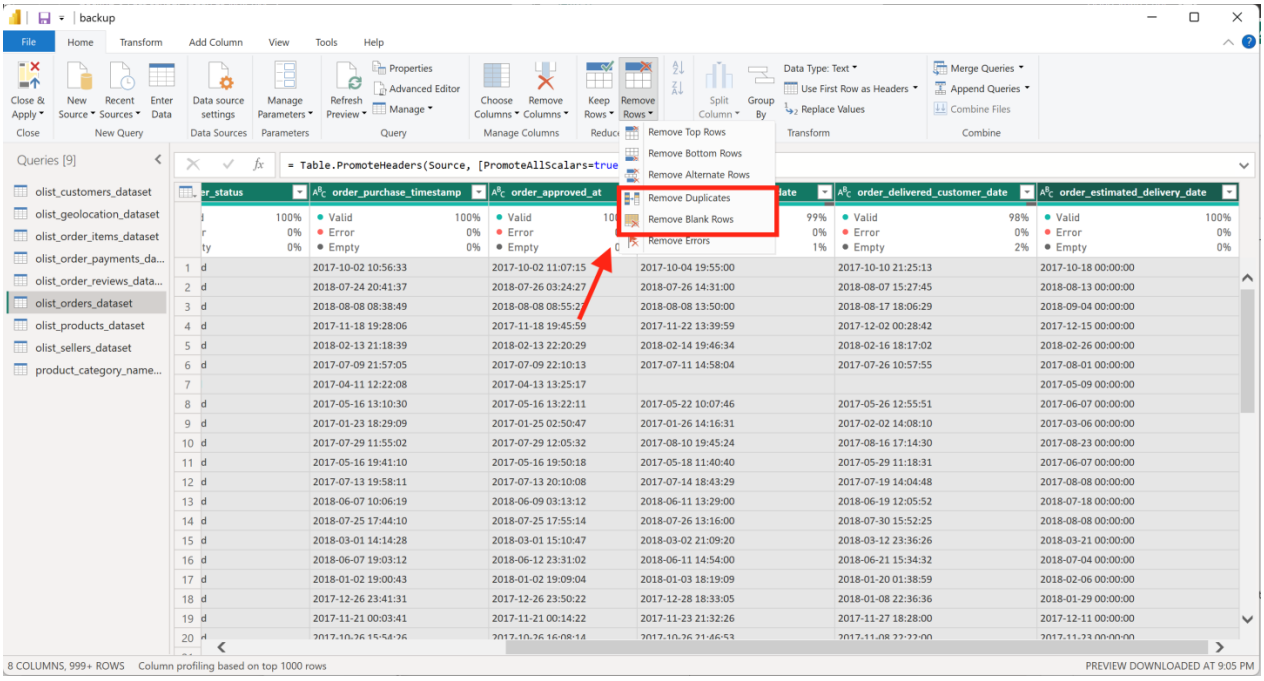
2.2.1 Data import



Picture 1:Import files into Power BI

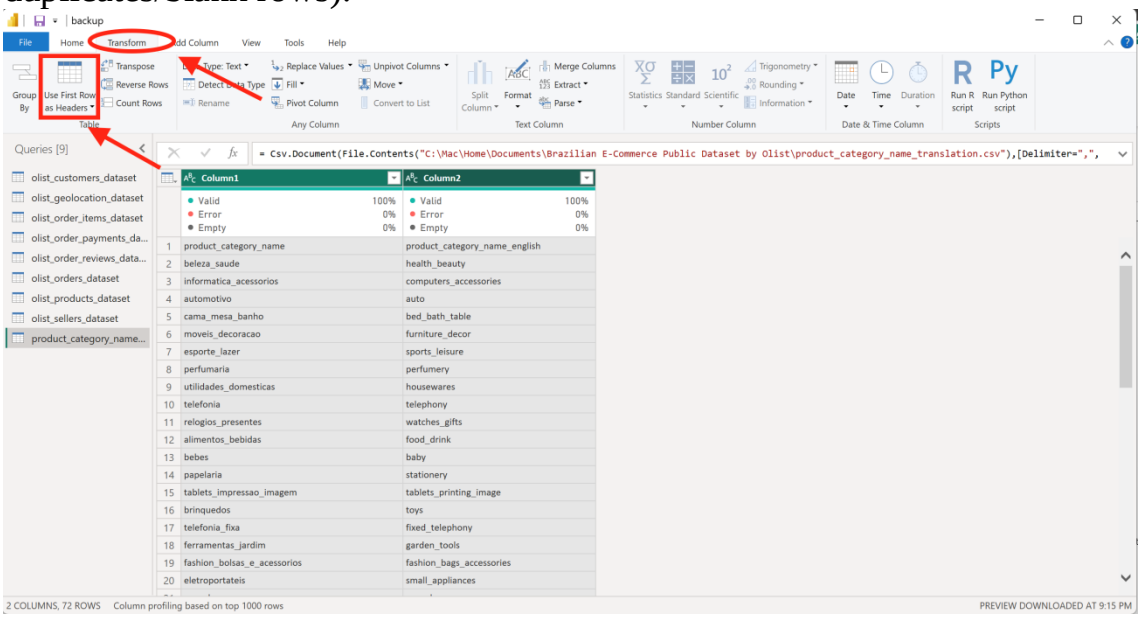
Import 9 files dataset into Power BI and click " Transform data" to open the Power Query Editor

2.2.2 Data cleaning



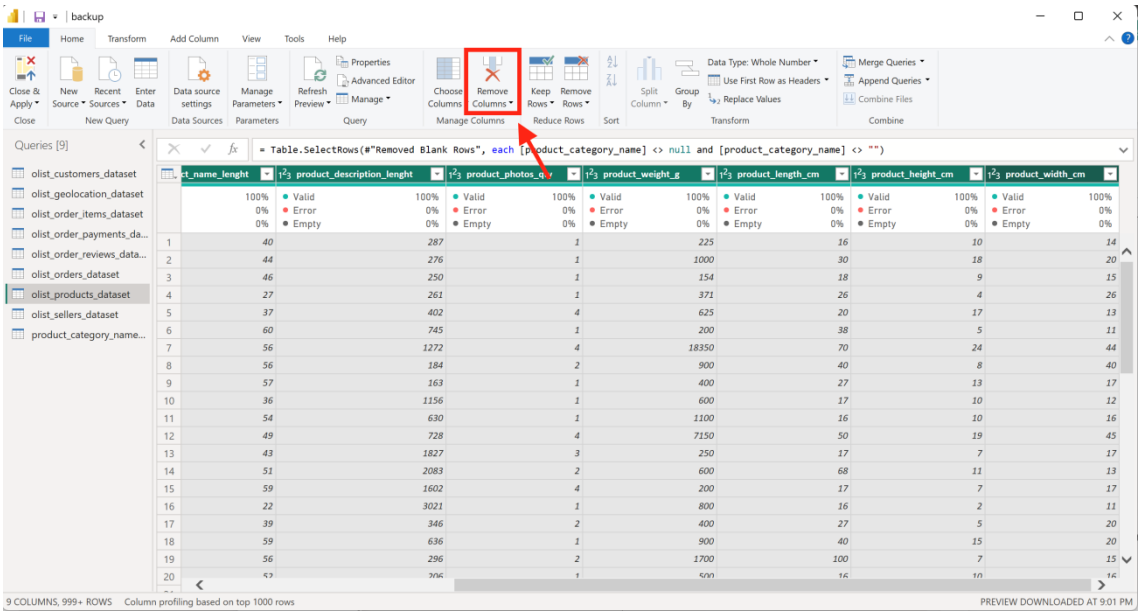
Picture 2: Remove duplicates and blank rows

The first step in the data cleaning process is to delete all rows with empty and duplicate values in the tables (open Remove rows -> Remove duplicates/blank rows).



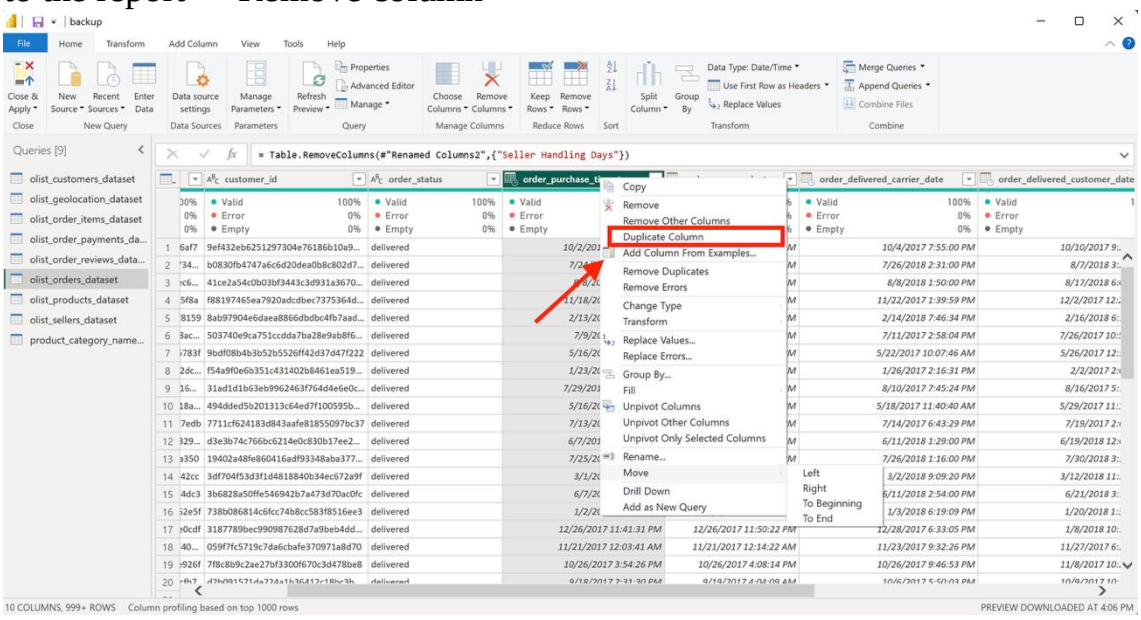
Picture 3: Use first row as header

Some data files often have errors where the first line is not used as a header. To fix this error, select Transform -> click on Use first row as header.



Picture 4: Delete unnecessary columns.

Removing unnecessary columns in Power BI optimizes model performance, reduces memory usage, minimizes analysis errors, and ensures data focuses on answering business questions. Data is only retained when it directly serves the purpose of analysis and decision-making. For example, fields like product length, weight, and number of images, etc., do not contribute to the report -> Remove column



Picture 5: Create OrderDate column

To create a time reference for the report, right-click on the order_purchase_timestamp column and select duplicate column to create an additional duplicate column as OrderDate. However, this is still a Date/Time data type, so we need to convert it to a Date data type.

2.2.3 Data modeling

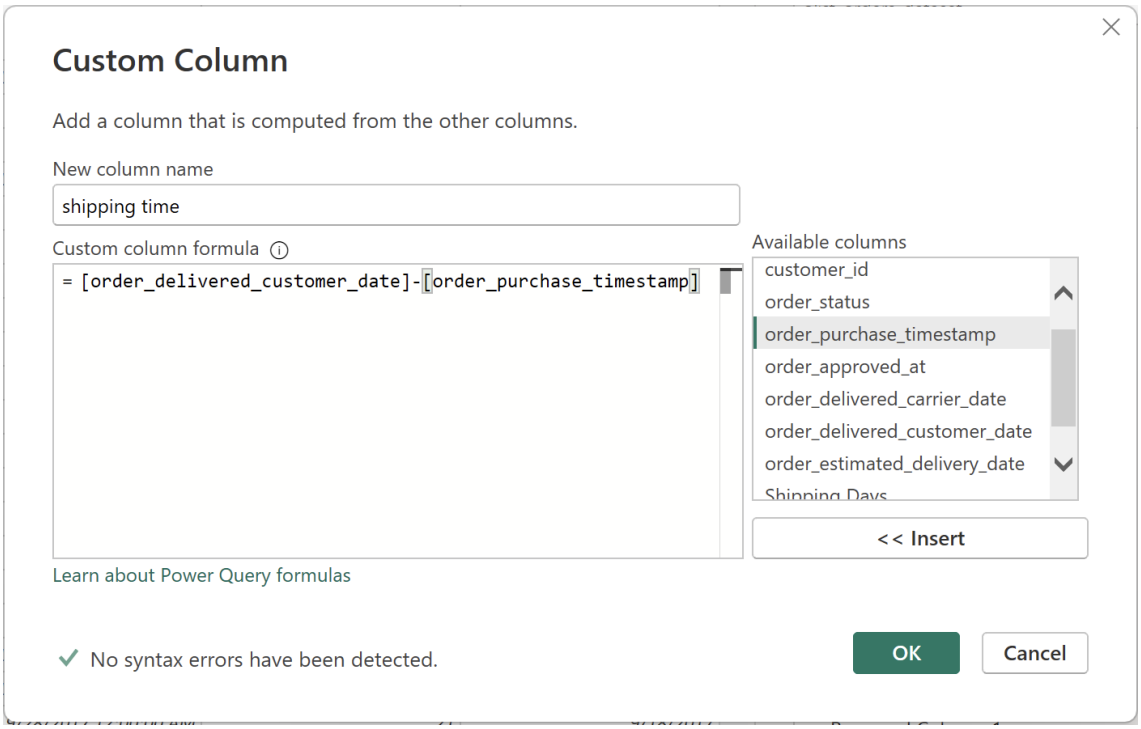
Picture 6: Data Model View

In data model, creating links between tables to ensure the data flow is on the right track.

- Order_items [order_id] to Order [order_id] (1-*)
- Order [order_id] to Reviews [order_id] (1-*)
- Order [order_id] to Payment [order_id] (1-*)
- Order [customer_id] to Customer [customer_id] (1-*)
- Order_items [product_id] to Products [product_id] (1-*)
- Order_items [seller_id] to Sellers [seller_id] (1-*)
- Create a new table in the model view with the formula DimDate =

ADDCOLUMNS(CALENDAR(DATE(2016,1,1), DATE(2018,12,31)),"Year", YEAR([Date]),"Month", FORMAT([Date], "MMMM"),"MonthNo", MONTH([Date]),"YearMonth", FORMAT([Date], "YYYY-MM"),"Quarter", "Q" & FORMAT([Date], "Q"),"Weekday", FORMAT([Date], "dddd"),"WeekdayNo", WEEKDAY([Date],2)).
 And link the date column to the OrderDate column in the olist_orders_dataset table to create a time reference.

2.2.4 Creation of Measures and Calculated Columns



Picture 7: Create shipping time column

In order_dataset table, Add custom column and use delivered_date column minus purchase_timestamp to calculate shipping time (need to convert to data type : Date)

```
Delivery Status =
IF(
    ISBLANK('olist_orders_dataset'[order_delivered_customer_date]),
    "delivered/canceled",
    IF(
        'olist_orders_dataset'[order_delivered_customer_date] > 'olist_orders_dataset'[order_estimated_delivery_date],
        "Late",
        "on time"
    )
)
```

Picture 8: Create delivery status column

To determine the on-time/late order status, go to Table View -> click on New Column and use the formula above to determine the order status.

```
Total Revenue = Sum (olist_order_items_dataset[price])
```

Picture 9: Use DAX measure to calculate Revenue

```
Revenue LY =
CALCULATE(
    [Total Revenue],
    SAMEPERIODLASTYEAR(DimDate[Date])
)
```

Picture 10: Use DAX measure to calculate Revenue Last Year

```
Revenue YoY =
[Total Revenue] - [Revenue LY]
```

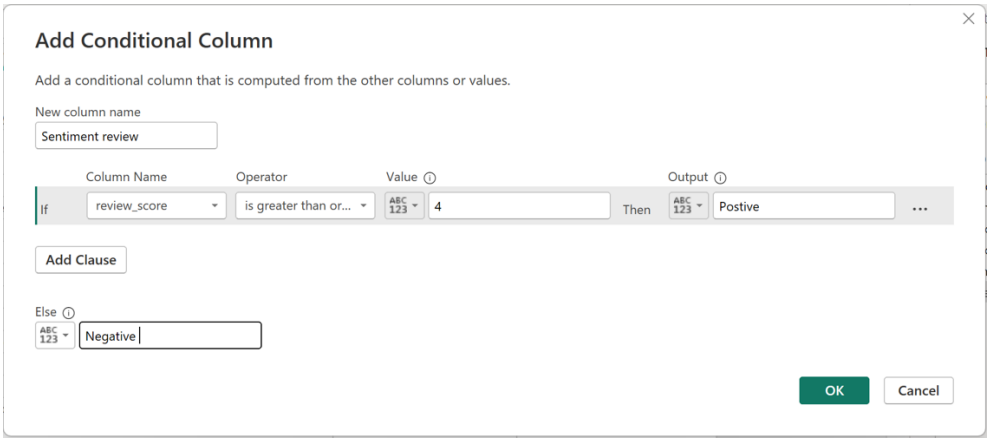
Picture 11: Use DAX mesure to calculate Revenue Year over Year

```
Late Orders =
CALCULATE(
    DISTINCTCOUNT(olist_orders_dataset[order_id]),
    olist_orders_dataset[Delivery Status] = "Late"
)
```

Picture 12: Use DAX measure to count Late order

```
Delay Rate =
DIVIDE(
    [Late Orders],
    [Total Orders],
    0
)
```

Picture 13: Use DAX measure to calculate Delay Rate



Picture 14: Add conditional column to seperate review score

In the review table, add a sentiment column to define customer ratings using positive (4-5 points) and negative (1-3 points).

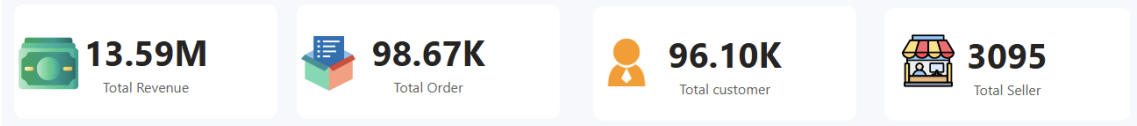
```
Seller Handoff Status =
VAR ActualCarrierDate = RELATED('olist_orders_dataset'[order_delivered_carrier_date])
VAR LimitDate = 'olist_order_items_dataset'[shipping_limit_date]
RETURN
IF(
    ISBLANK(ActualCarrierDate),
    "delivered/canceled",
    IF(ActualCarrierDate > LimitDate, "seller late", "seller on time")
)
```

Picture 15: Create column to define seller status

In the order_item table, create an additional column with the formula above to determine which sellers deliver on time and which deliver late.

3. Evaluation and Analysis

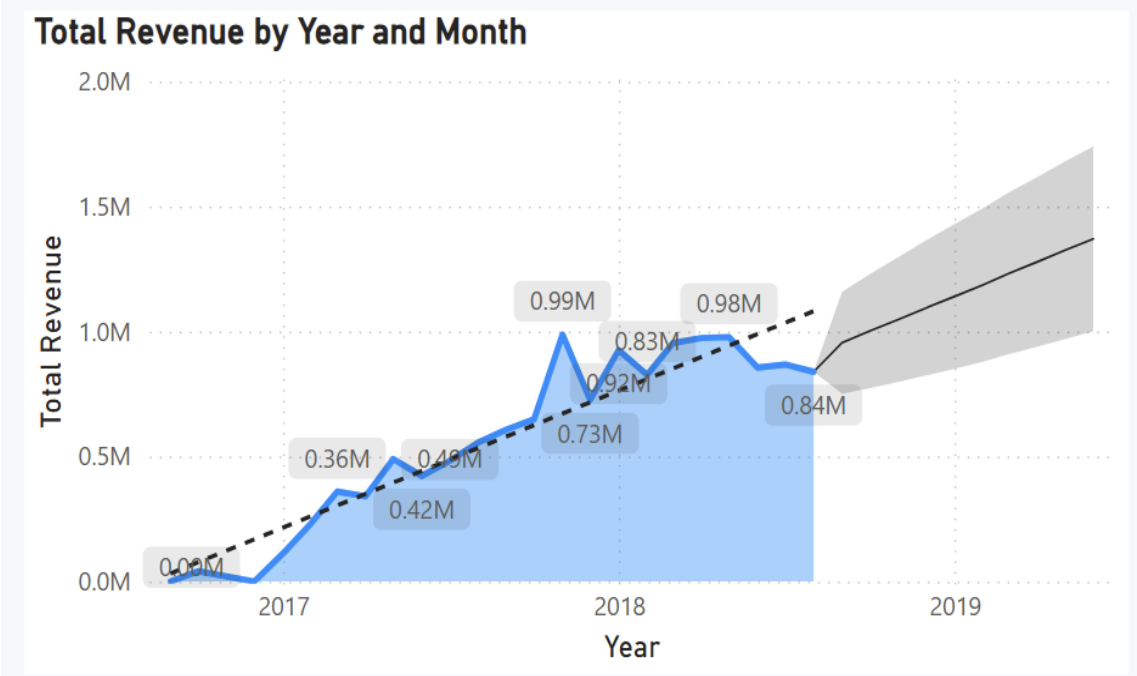
3.1 Sales overview



Picture 16: Overview about Olist

Olist is an e-commerce platform established in 2015. Between 2016 and 2018, the platform generated a total of 98.67K orders, 96.10K

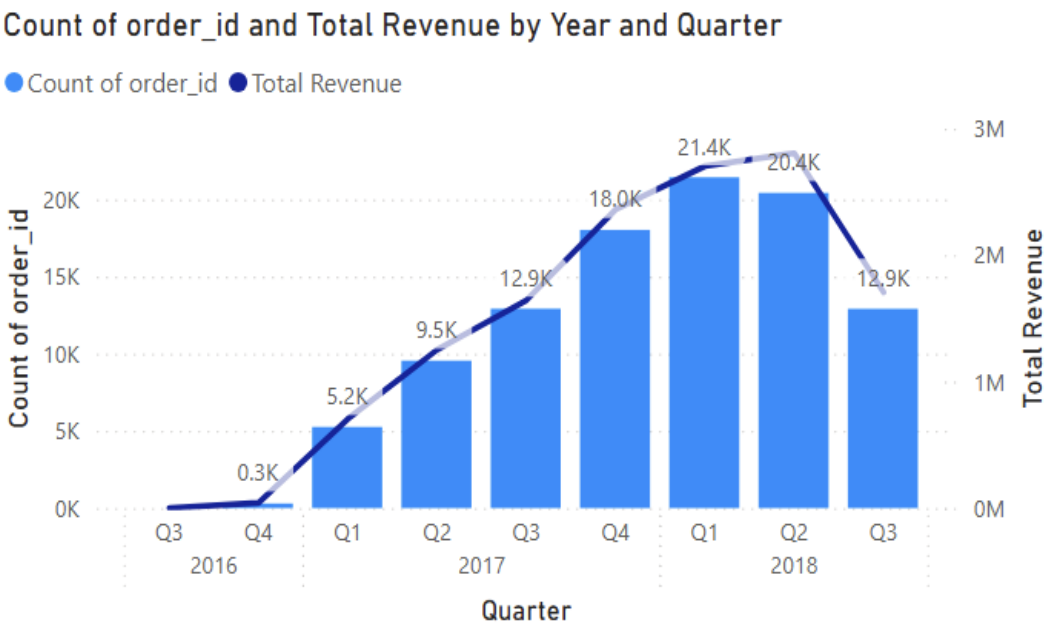
customers, and 3095 sellers. During the same period, the platform's revenue reached $13.59 million, with an annual growth rate of 1.26.



Picture 17: Line chart show total revene by year/month

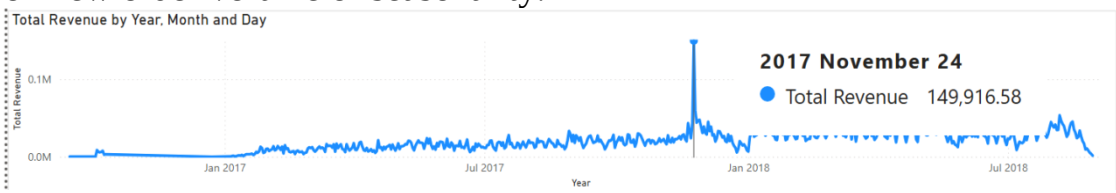
The dashed black line shows a clear upward growth trend from the beginning of 2017 to mid-2018. This demonstrates that the Olist e-commerce platform was in a period of strong expansion and attracting a large number of new users. Revenue peaked around the end of 2017 at 0.99M. According to Olist's actual data, this was the impact of the Black Friday event in November 2017. After this surge, revenue experienced a slight decrease (to 0.73M) but immediately established a new revenue base significantly higher than the previous period. In 2018, revenue no longer increased sharply but began to fluctuate steadily between 0.8M and 1.0M per month. This indicates that the business has passed the initial phase of entering the hot market and is entering a phase of maintaining a stable market share.

Growth potential: The shaded area at the bottom of the chart is Power BI's Forecast feature, predicting continued revenue growth in 2019.



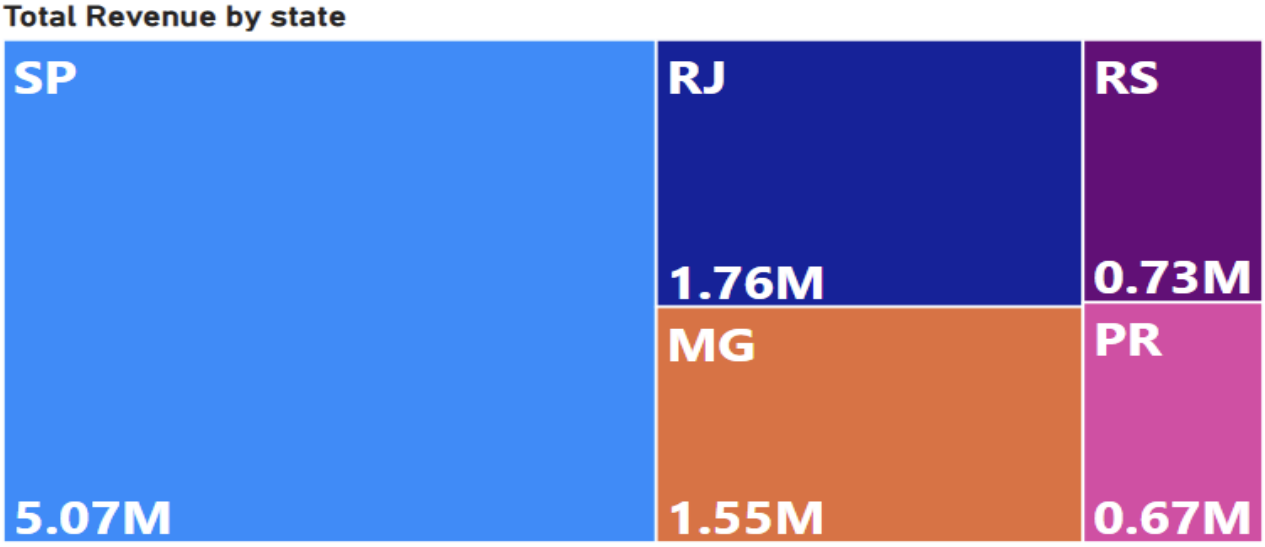
Picture 18: Line and clustered column chart show total order/ revenue by quarter

From Q1/2017 (5.2K orders) to Q4/2017 (18.0K orders), the number of orders grew by over 246% in just one year. This was a period of extremely successful market entry for Olist. The number of orders peaked in Q1/2018 with 21.4K orders. This is often related to post-Tet (Lunar New Year) promotions or early-year shopping campaigns. There was a sharp decline in both orders and revenue in Q3/2018 (down to 12.9K orders). High covariance: The line (Revenue) and the column (Number of Orders) move almost perfectly parallel. This indicates that Olist's Average Order Value (AOV) is quite stable across quarters, without significant fluctuations in product mix. Since revenue depends on the number of orders, Olist's current strategy focuses on generating more new orders rather than the value of each order. Looking at the line graph, it's clear that Olist's revenue growth phase from the previous year has begun to slow down. Continuing to overemphasize the number of new orders could lead to increased marketing costs or unforeseen, unpredictable factors. Instead, Olist needs to incorporate upsell/cross-sell strategies to increase the value of each order, thereby achieving sustainable revenue growth that isn't overly dependent on new order volume or seasonality.



Picture 19: Line chart show total revenue by day

When analyzing daily sales figures throughout the year, the line graph shows that there was one peak day that significantly outperformed the others during the period from 2016-2018: November 27, 2017 (~$150K), coinciding with Black Friday week. This indicates that customers typically have high demand for goods during holidays like Black Friday.
->Therefore, to optimize sales during peak season, businesses need to start marketing campaigns and negotiate with sellers at least from Q3 to prepare inventory and offers for this period. Establishing a "Pre-Black Friday" program can alleviate logistics pressure and capitalize on early customer demand.



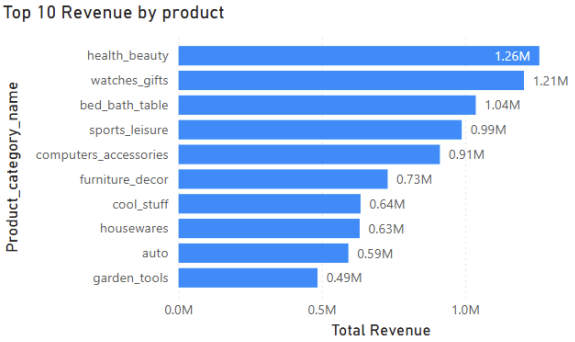
Picture 20: Tree map shows total revenue by state

The treemap chart of the top 5 revenue-generating states reveals an extremely high concentration of the market. São Paulo (SP) holds a dominant position with 5.07M in revenue, contributing nearly three times more than the second-ranked state, Rio de Janeiro (1.76M). This reflects the fact that SP is the economic and logistics hub of Brazil. These three states (São Paulo, Rio de Janeiro, Minas Gerais) account for an overwhelming share of total revenue. This region has the most developed transportation infrastructure, allowing for faster delivery times and lower shipping costs.
-> With nearly 50% of revenue coming from São Paulo, any disruption in this state (such as strikes or warehousing issues) would severely damage Olist. A separate logistics contingency plan is needed for this region.

-> Exploiting Niche Markets in Northern/Northeastern States: Revenue in states outside the Top 5 is very low. Olist could implement a "Flat-Rate Shipping" campaign or recruit more sellers in those areas to reduce shipping distances and stimulate demand.

3.2 Products

- **Top 10 highest-selling products at Olist Store**



product_category_name_english	Revenue	Avg review score	Freight Ratio	Avg Delivery time
health_beauty	1,258,681.34	3.62	0.15	11.61
watches_gifts	1,205,005.68	3.31	0.08	12.33
bed_bath_table	1,036,988.68	3.47	0.20	12.52
sports_leisure	988,048.97	3.55	0.17	11.78
computers_accessories	911,954.32	3.38	0.16	12.73
furniture_decor	729,762.49	3.34	0.24	12.59
cool_stuff	635,290.85	3.12	0.13	11.92
housewares	632,248.66	3.38	0.23	10.62
auto	592,720.11	3.12	0.16	11.87
garden_tools	485,256.46	3.08	0.20	13.16

Picture 21: Bar chart and matrix table show top 10 highest revenue by product and Avg review score/ delivery time

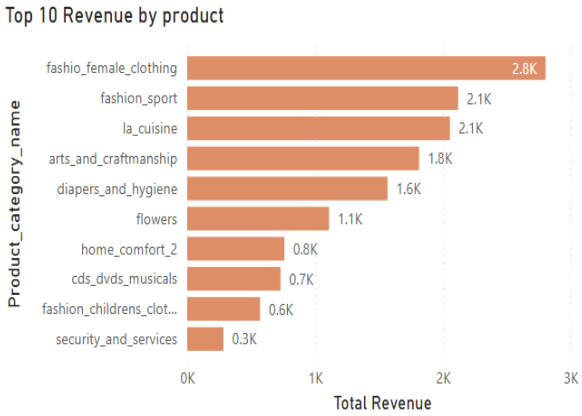
Health_beauty and bed_bath_table are the two leading product categories (accounting for ~40% of the top 10), reflecting high demand for personal and household products in Brazil – a rapidly urbanizing market. Olist should prioritize partnering with sellers specializing in these items to increase inventory and reduce prices, capitalizing on the health and beauty trend.

Categories like watches_gifts (0.08) and computers_accessories (0.16) have low shipping costs, helping Olist compete with other competitors. These top-selling items, combined with low shipping costs relative to price, are also perfectly suited for upsell/cross-sell strategies as mentioned above.

In this product group, the average rating is neither too high nor too low; however, garden_tools (3.08) and cool_stuff (3.12) have lower ratings, possibly due to poor product durability. Olist could create more promotional programs and pay attention to the packaging and shipping process for this product group to achieve higher ratings.

The average delivery time for an order in this category is 11.9 days; however, for furniture_decor items, it reaches 12.59 days. This could be due to the volume or size of the items slowing down the overall delivery process. -> Olist needs to add more local warehouses, especially in areas with high order volumes such as São Paulo, Rio de Janeiro, etc.

- **Top 10 lowest – selling producs at Olist store**



product_category_name_english	Revenue	Avg review score	Freight Ratio	Avg Delivery time
fashio_female_clothing	2,803.64	1.79	0.22	12.42
fashion_sport	2,119.51	1.79	0.27	10.15
la_cuisine	2,054.99	4.25	0.16	7.23
arts_and_craftmanship	1,814.01	4.17	0.20	5.39
diapers_and_hygiene	1,567.59	1.78	0.37	9.96
flowers	1,110.04	4.39	0.44	11.03
home_comfort_2	760.27	3.83	0.54	14.00
cds_dvds_musicals	730.00	4.67	0.31	10.50
fashion_childrens_clothes	569.85	1.77	0.17	8.14
security_and_services	283.29	2.50	0.15	15.00

Picture 22: Bar chart and matrix table show top 10 lowest revenue by product and Avg review score/shipping time

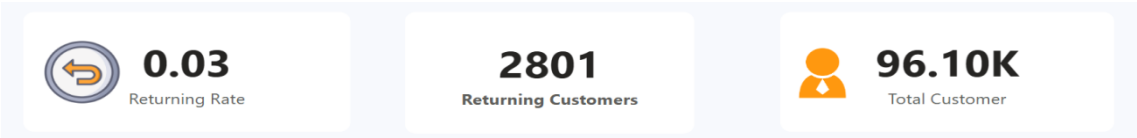
The fashion category (female, sport, children's) was a clear failure: accounting for 3 out of the top 10 lowest-selling categories, with record-low reviews (1.77–1.79) – indicating poor quality and unsatisfactory customer experience. Olist may remove or tightly control its sellers to avoid impacting the overall ranking.

Items with high freight ratios: ome_comfort_2 (0.54) and flowers (0.44) have shipping costs accounting for half of their revenue – bulky/perishable items. Olist should partner with logistics companies specializing in shipping these types of items to reduce shipping costs and increase revenue.

High reviews but low sales: Cds_dvds_musicals (4.67), flowers (4.39), la_cuisine (4.25) have excellent reviews but low sales, possibly due to the small market size of these items. However, the impressive reviews demonstrate a very promising opportunity for investment in marketing or products. Olist could experiment with boosting marketing during seasons/holidays and collaborating with relevant parties for a specific period to assess sales results from these items, and pay particular attention to shipping for items like flowers.

3.3 Customer

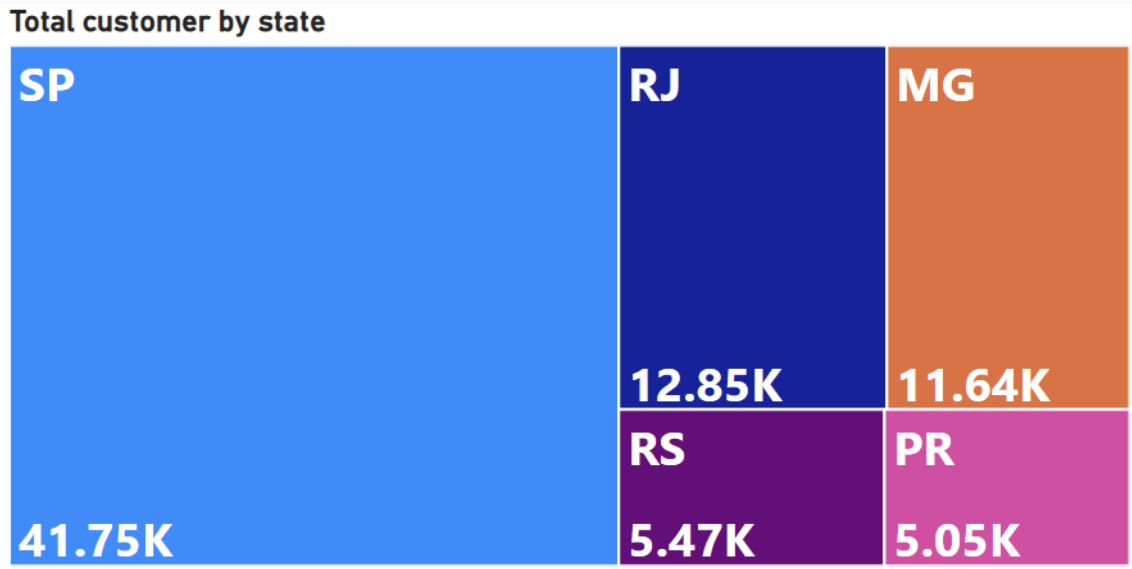
- **Customer overview**



Picture 23: Customer overview

With a total customer base of nearly 100,000 but a return rate of only 3%, coupled with the revenue chart analysis from the previous section, this further demonstrates that Olist is primarily operating as a marketplace catering to one-time purchases rather than a platform for creating repeat purchases. This results in extremely low customer lifetime value and current revenue growth heavily reliant on acquisitions. Consequently, when Olist's prices are less attractive than those of other platforms, it fails to attract new customers, leading to significant losses in the platform's revenue.

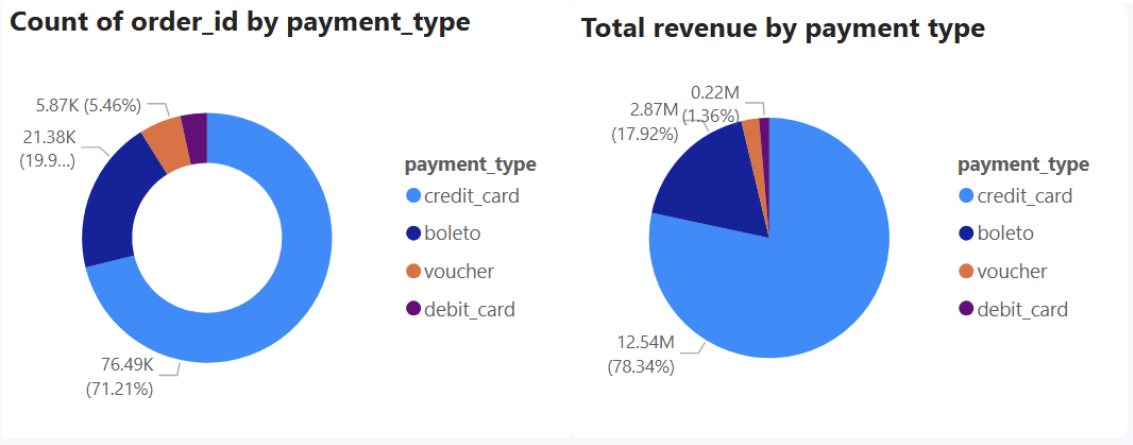
- **Top 5 states with the most customers**



Picture 24: Tree map shows the amount of customerby state

The number of customers by state is also similar to the revenue by state, with São Paulo (41.75K), Rio de Janeiro (12.85K), Minas Gerais (11.64K), Rio Grande do Sul (5.47K), and Paraná (5.05K) respectively.

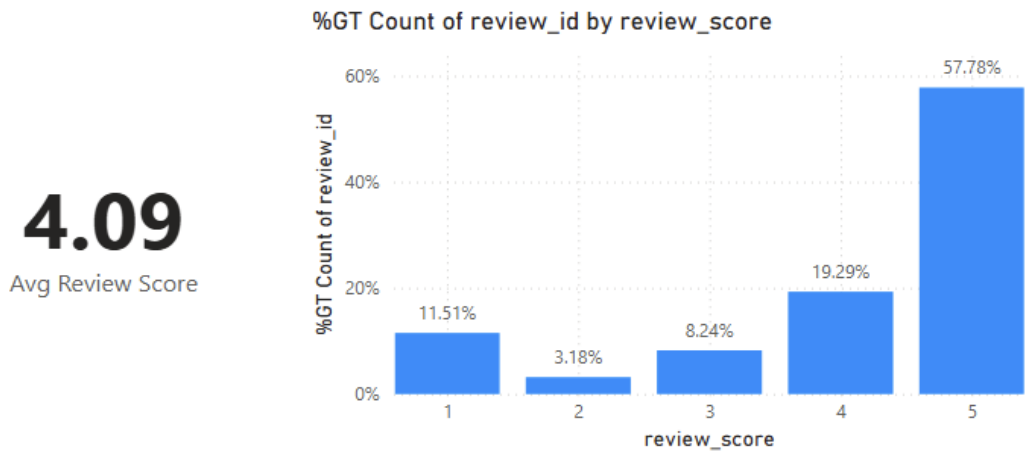
- **Favourite payment type**



Picture 25: Donut chart and Pie chart show total order/ revenue by payment_type

There are four main payment methods on Olist: credit card, boleto (a payment voucher for those without a bank account), voucher, and debit card. Credit cards dominate both in terms of volume and revenue, indicating that Olist customers prioritize quick and convenient payments. Furthermore, credit cards allow installment payments, further stimulating demand for high-value items. Next is boleto, accounting for 19.92% of orders but only 17.92% of revenue – meaning the average order value is lower. Boleto's popularity stems from its suitability for those without bank accounts or who prefer not to share personal financial information. Methods like vouchers and debit cards are generally more popular for low-value items.

- **Review score**

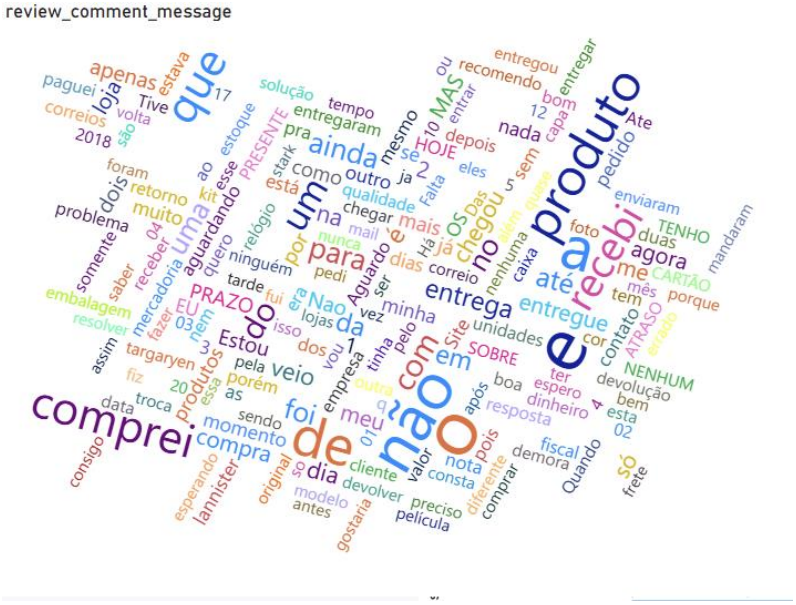


Picture 26: Average review score and the amount review of customer by review score

The average customer rating is 4.09/5, an incredibly impressive number. Looking deeper, we see that positive reviews (> 4 points) account for over 70%, while negative reviews are quite low at 8.24% (3 points) and

3.18% (2 points). However, there is a noteworthy 11.51% rating of 1 point.

- **Customer reviews give it 1 star**


review_comment_message
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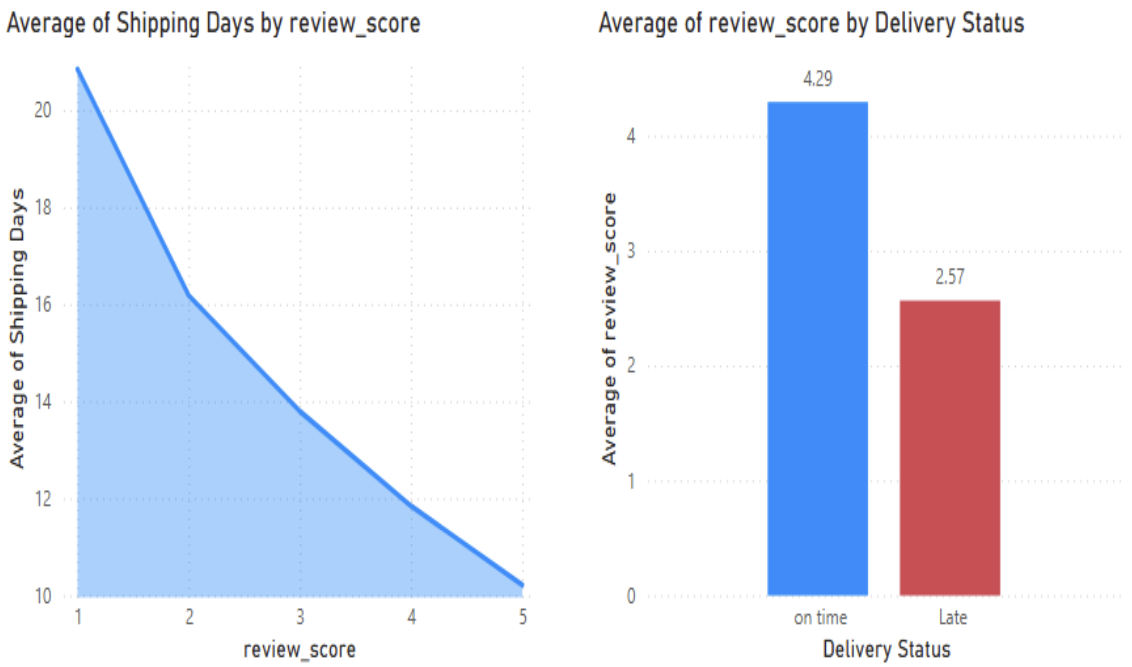
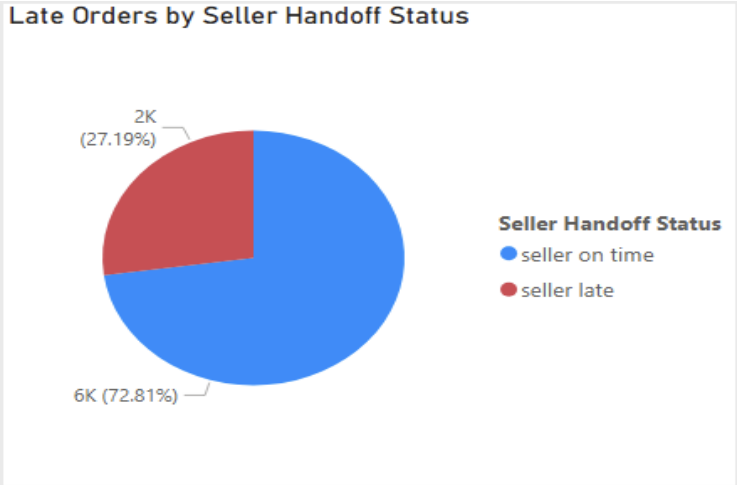


Picture 28: Line chart and column chart show the correlation between Avg shipping time and Avg review score

When investigating the factors influencing ratings, the line graph shows a correlation between delivery time and average score: the higher the delivery time, the lower the score.
Furthermore, looking at the bar chart, it can be seen that orders delivered on time generally receive a positive rating of 4.29 points, while orders delivered late receive only 2.57 points.

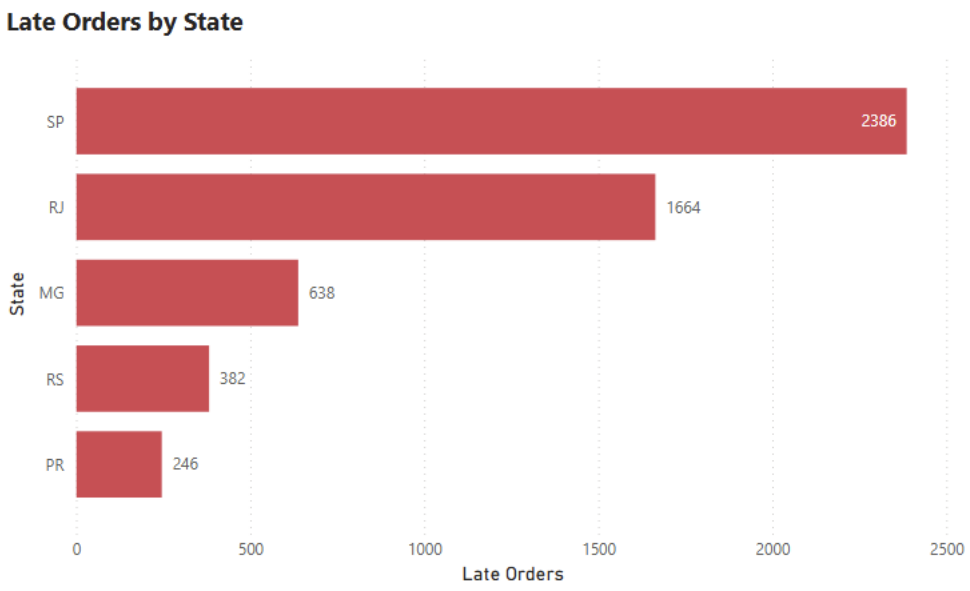
- **Identify the cause of the delivery delay.**



Picture 29: Pie chart shows the Seller's delivery status on delayed orders

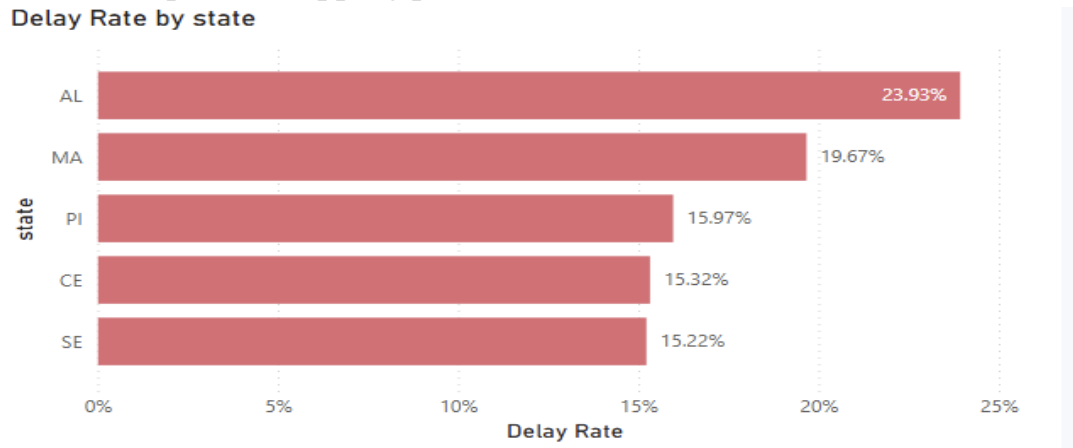
When analyzing delayed orders, I found that 72.81% of sellers delivered on time, while only 27.19% delivered late. This suggests that the majority of delayed deliveries are due to the shipping company.

- **5 states with highest late orders**

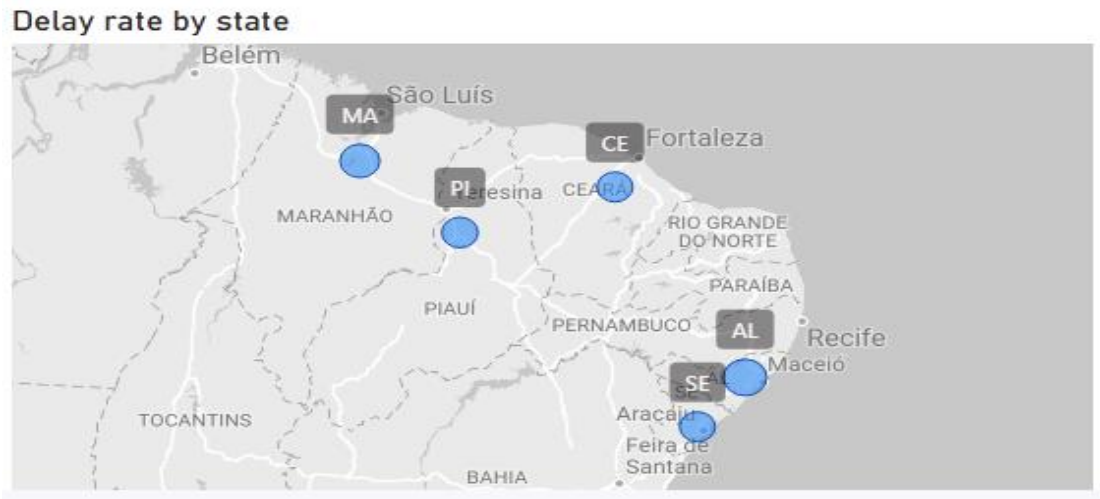


Picture 30: Bar chart shows 5 states with highest late orders

The chart above reflects the 5 states with the highest number of delayed orders. Notably, these are also the states with the highest number of orders, revenue, and customers. Customers in these states shop online extensively (Brazilian e-commerce is concentrated in urban areas), leading to high order/revenue ratios and consequently high late orders. This indicates that Olist is successful in its core market but has not yet optimized its operations. Olist could potentially lose more than 50% of its revenue if an unannounced event disrupts the shipping process to these states.



Picture 31: Bar chart shows 5 states with highest delay rate



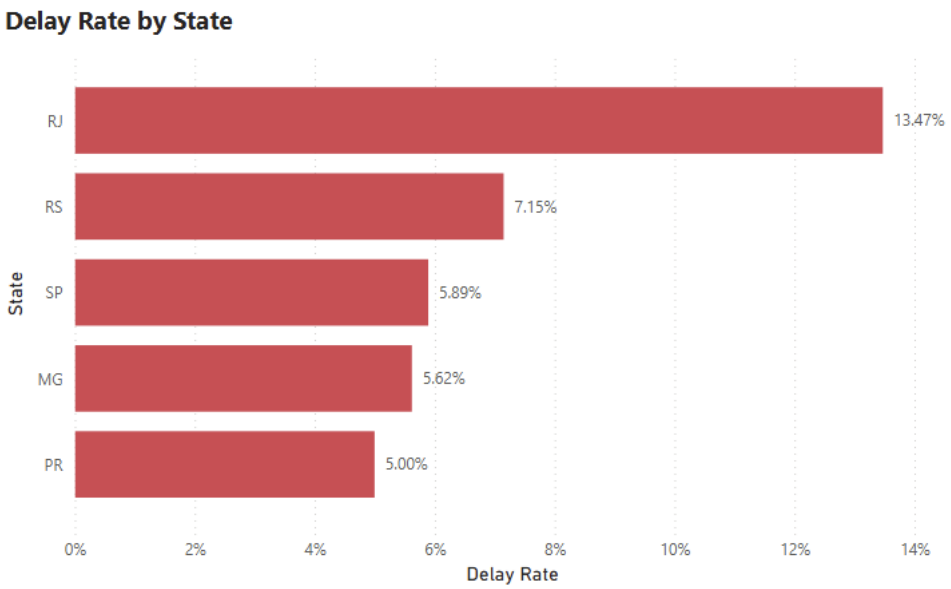
Picture 32:Geographic location of the states with the highest delay rates:

When analyzing the rate of delayed deliveries, the five states with the highest delay rates are Algoas (23.93%), Maranhão (19.67%), Piauí (15.97%), Ceará (15.32%), and Sergipe (15.22%). A common factor among these five states is their location in the Nordeste region (Northeast Brazil). Delivery to the Nordeste region (Northeast Brazil) is often delayed due to a combination of factors:

- **Limited and degraded road infrastructure**
  - Low road density: Although Nordeste has a fairly long federal road network, the road density is only 12.5 km per 1,000 km², significantly lower than the southern region (20.5 km). This limits the ability to connect quickly between localities.
  - Poor road quality: A large portion of roads in Nordeste are classified as "Regular," "Ruim," or "Péssimo." Poor road conditions force drivers to reduce speed, causing wear and tear on vehicles and extending travel times.

  - High percentage of unpaved roads: Approximately 78.5% of the national road network remains unpaved, leading to slow and difficult travel, especially in the inland regions of Nordeste. (Costa et al., n.d.)
- **Olist operational characteristics**
  - High average delivery time: According to Olist's data analysis, the average delivery time is 12.09 days, but in Nordeste and remote areas, this figure is often much higher due to the

geographical distance from the main warehouses (typically concentrated in São Paulo and Rio de Janeiro).



Picture 33: Late pregnancy rates in the five central states of Olist.

While the highest number of delayed orders was in Sao Paulo, Rio de Janeiro had the highest rate of late applications among the top five most prominent states at 13.47%. The reason for this high delay rate is as follows:

- Challenging Terrain: The city's diverse geography and challenging terrain make transportation inherently more complex and costly.
- Severe Congestion: Rio de Janeiro is ranked as one of the world's 10 worst cities for traffic, with road traffic volume growing by over 40% in the last decade.
- Inadequate Transport Networks: Many regions lack sufficient connectivity, and poor road conditions lead to delays and high maintenance costs.

3.4 Seller

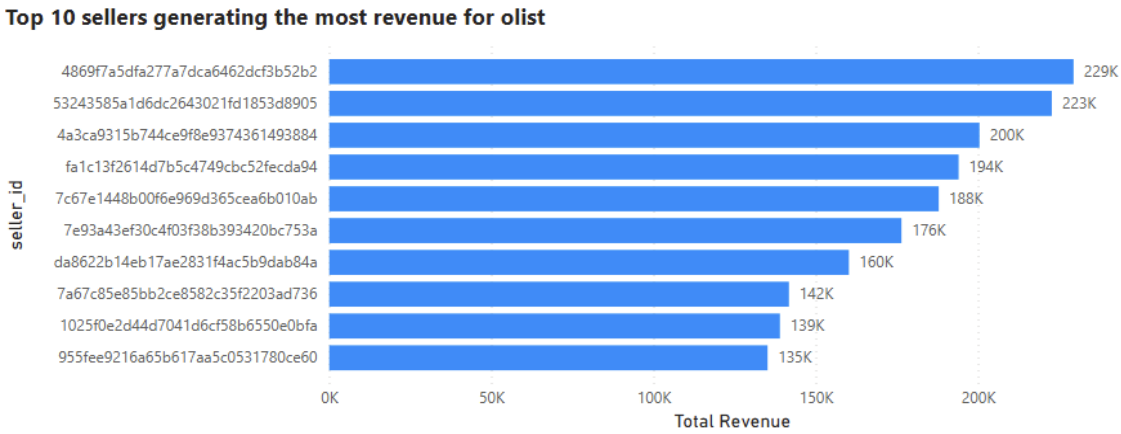
- **10 sellers with the most delayed**

seller_id	Late Orders	Average of review_score
4a3ca9315b744ce9f8e9374361493884	195	2.41
1f50f920176fa81dab994f9023523100	148	2.50
4869f7a5dfa277a7dca6462dcf3b52b2	130	2.23
ea8482cd71df3c1969d7b9473ff13abc	118	2.83
6560211a19b47992c3666cc44a7e94c0	117	2.62
cc419e0650a3c5ba77189a1882b7556a	101	2.51
da8622b14eb17ae2831f4ac5b9dab84a	100	2.62
7c67e1448b00f6e969d365cea6b010ab	98	2.13
955fee9216a65b617aa5c0531780ce60	98	2.91
8b321bb669392f5163d04c59e235e066	97	2.48

Picture 34: Matrix table shows 10 sellers have highest late orders

When creating a matrix of the 10 sellers with the most delayed deliveries, their average customer ratings were extremely low, ranging from 2.23 to 2.91. -> Olist needs to address or even remove these sellers to avoid negatively impacting the customer experience.

- **Top 10 sellers generating the highest revenue for Olist**



Picture 35: Bar chart shows 10 sellers generating the most revenue for Olist

seller_id	Total Revenue	Delay Rate
4869f7a5dfa277a7dca6462dcf3b52b2	229,472.63	0.12
53243585a1d6dc2643021fd1853d8905	222,776.05	0.04
4a3ca9315b744ce9f8e9374361493884	200,472.92	0.11
fa1c13f2614d7b5c4749cbc52fecda94	194,042.03	0.10
7c67e1448b00f6e969d365cea6b010ab	187,923.89	0.10
7e93a43ef30c4f03f38b393420bc753a	176,431.87	0.06
da8622b14eb17ae2831f4ac5b9dab84a	160,236.57	0.08
7a67c85e85bb2ce8582c35f2203ad736	141,745.53	0.06
1025f0e2d44d7041d6cf58b6550e0bfa	138,968.55	0.10
955fee9216a65b617aa5c0531780ce60	135,171.70	0.08

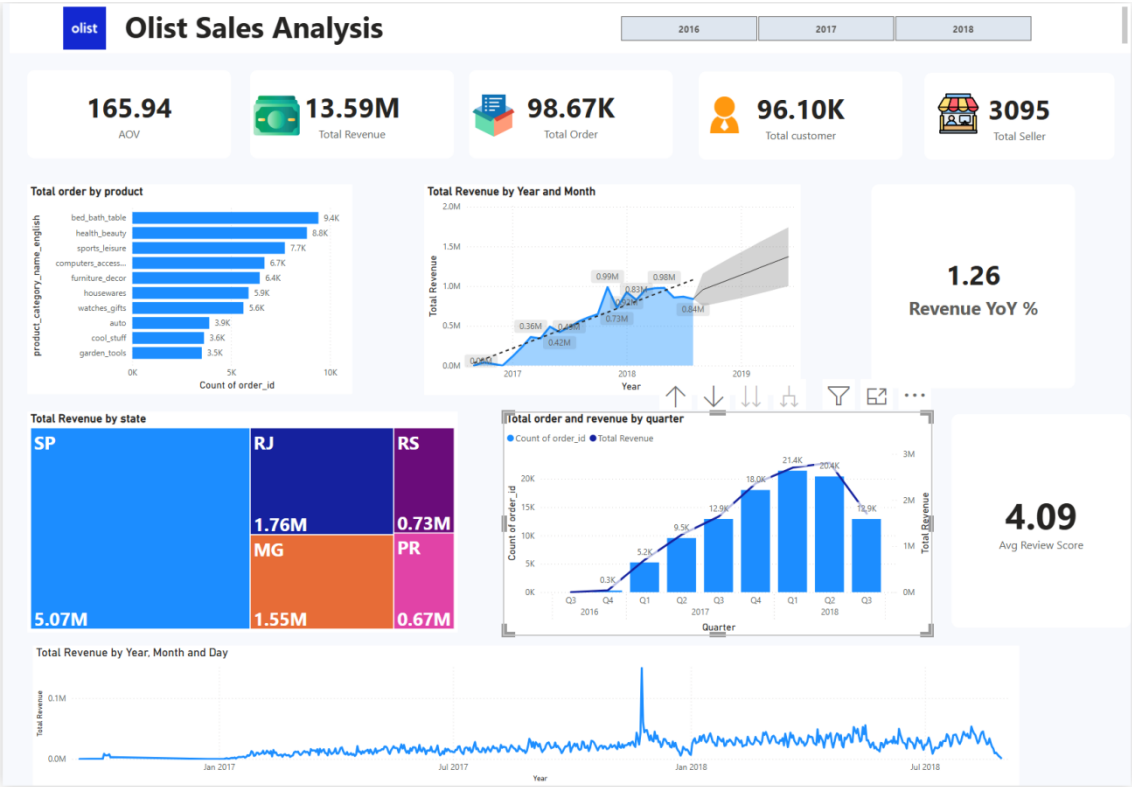
Picture 36: The matrix table shows revenue and delay rate of seller

Above are the top 10 highest-earning sellers for Olist during the period from 2016-2018, notably 9 out of 10 sellers are located in Sao Paulo. A deeper analysis reveals that these sellers all have very low late delivery rates (<15%). Besides product quality, the customer experience they provide also contributes to increased revenue. Olist needs strategies to retain these sellers in order to sustainably develop its e-commerce platform.

4. Conclusion and Recommendations

4.1 Conclusion

- **Revenue and Growtth**



Picture 37: Sales Dashboard

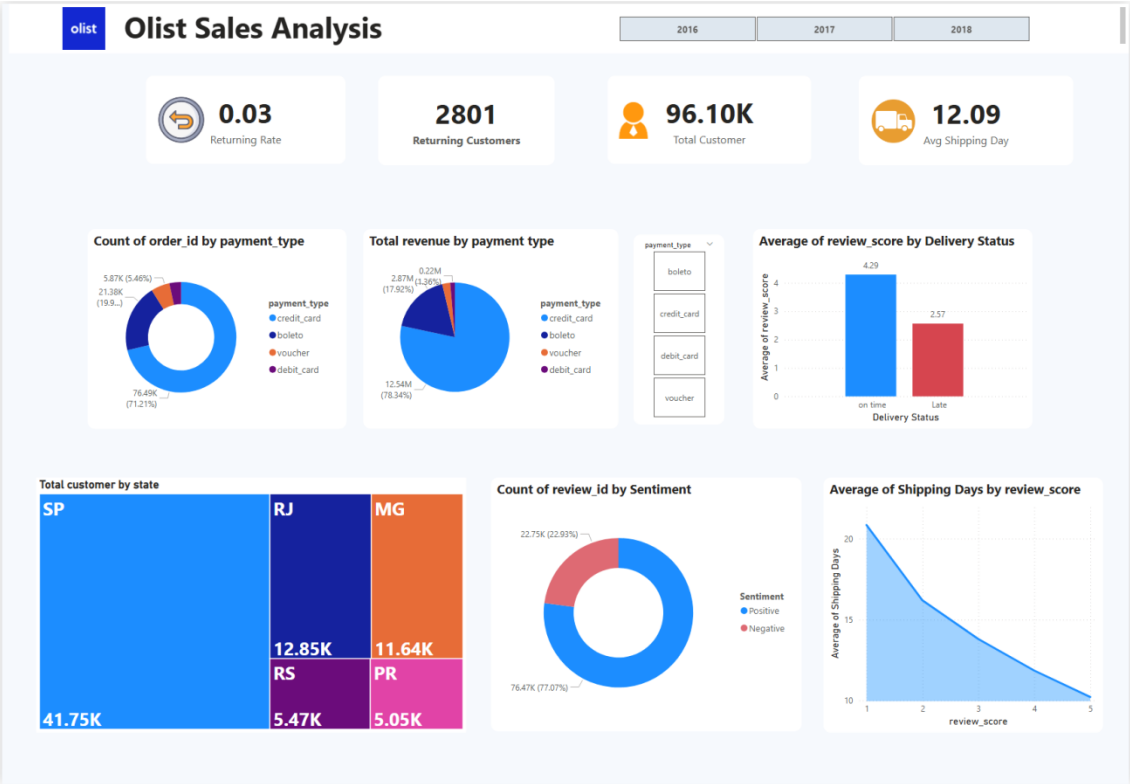
Revenue increased sharply from 2017 to mid-2018, peaking in November 2017, with a record high of 21,400 orders in Q1 2018. Growth slowed after 2018, indicating market saturation or the need for a new strategy.
Olist's revenue is primarily concentrated in three states: São Paulo, Rio de Janeiro, and Minas Gerais

- **Products**

Best-selling products: Health & beauty, bed & bath, watches & gifts

Low-selling products: Fashion (women's, sportswear, children's) have low ratings; consider removing or improving the quality and seller awareness for these items.
Opportunities: Some categories like flowers, CDs/DVDs have high ratings but low sales.

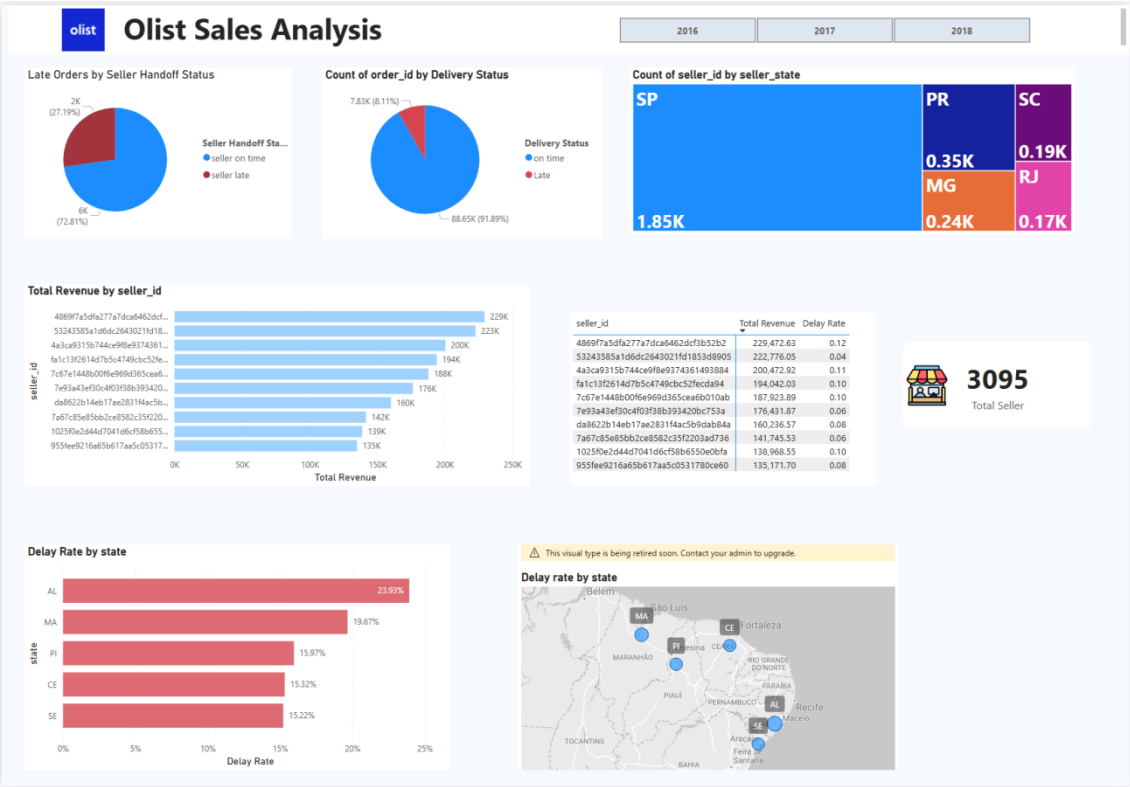
- **Customer**



Picture 38: Customer Dashboard

With a customer return rate of only 3%, the platform's main customer base consists of new customers who haven't yet established a strong presence in the market.
Customers prefer using credit cards, with 74% of payments made via this method.
The average rating of 4.09/5 is quite good, but a deeper analysis of the ratings reveals that 11.05% of customers gave a 1-star rating (mostly due to shipping issues).
There is a correlation between delivery time and customer ratings (longer delivery times result in lower ratings).

- **Seller**



Picture 39: Seller Dashboard

27.19% of delayed orders were due to the seller, but the majority were caused by the shipping company.
Some sellers had high rates of delayed deliveries, resulting in low ratings → need monitoring or removal.

- The area with the most delayed deliveries was Alagoas (23.93%) due to its degraded road infrastructure

4.2 Recommendations

- The graph of revenue versus the number of orders shows that olist's revenue increased due to the number of orders, not the value per order. Olist needs to improve product quality to increase value per order, thereby achieving sustainable growth.
- Olist needs to prepare inventory and shipping for holidays to meet customer demand during these occasions.
- Upsell/crosssell strategies should be added for high-revenue items such as Health_beauty and watches_gift.
- Pilot seasonal marketing programs with low-revenue but high-rated products like CDs_dvds_musicals (4.67), flowers (4.39), and la_cuisine (4.25) to fully exploit the potential of these items.

- Remove or remind sellers of products with both low revenue and low ratings, such as fashion categories (female, sport, children's).
- A customer return rate of only 3% shows that Olist is heavily reliant on new customers. With the market saturated, Olist needs to reposition its target customer group. (This refers to a loyal customer group). Olist needs to have customer loyalty programs or develop targeted marketing campaigns for high-risk customer segments using appropriate promotions, re-engagement emails, and product recommendations based on previous purchase history.
- With credit card payment being the preferred method, Olist should partner with banks to organize programs such as 20% cashback on high-value products when paying by credit card or installment plans to attract customers.
- Open more warehouses and delivery points in states such as São Paulo, Rio de Janeiro, and Minas Gerais to avoid order overload.
- Expand delivery capacity: Invest in partnerships with logistics providers to reduce delivery times, especially in areas experiencing the longest delays. This is necessary to both improve customer satisfaction and retain existing customers.
- There is a need for incentive policies for Nordester employees and salespeople who generate high sales for the platform

5. References

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